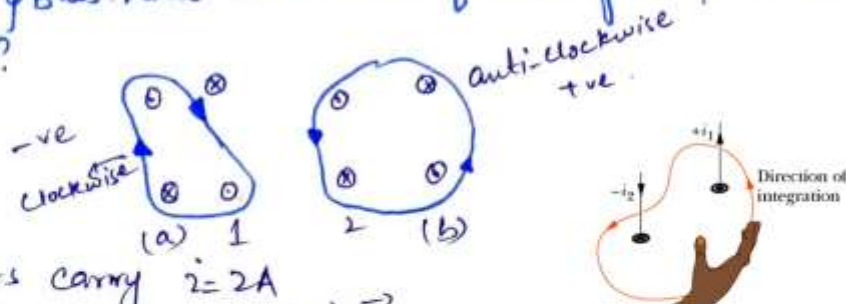


P-45: Each of the eight conductors in Fig 29-69 carries 2.0 A of current into or out of the page. Two paths are indicated for the line integral $\oint \vec{B} \cdot d\vec{s}$. What is the value of the integral for (a) Path 1 and (b) Path 2?



8 conductors carry $i = 2A$

Two paths indicated for $\oint \vec{B} \cdot d\vec{s}$

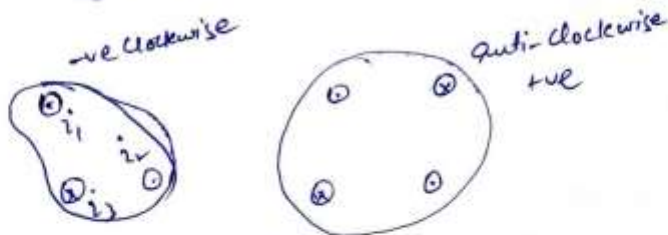
What is the value of the Integral (a) Path 1
(b) Path 2

⊗ path is in clockwise (current into the page $\rightarrow +ve$
R.H.R (Ampere's law) and ○ current out of the page $\rightarrow -ve$

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_2$$

$$= -\mu_0 i_2 = -4\pi \times 10^{-7} (2)$$

$$\oint_a \vec{B} \cdot d\vec{s} = -2.5 \times 10^{-6} Tm$$



i_1 and i_2 will cancel out

$$i = i_2 = i_3 = 2A$$

$$(b) \oint_b \vec{B} \cdot d\vec{s} = 0$$

two currents are out of the page and two currents are into the page.

P-43: Figure 29-67 shows a cross section across a diameter of a long cylindrical conductor of radius $a = 2 \text{ cm}$ carrying uniform current 170 A . What is the Magnitude of the current's Magnetic field at radial distance (a) 0, (b) 1.00 cm , (c) 2.00 cm (wire's surface), and (d) 4.00 cm ?

$a = 2 \text{ cm}$ of long cylindrical conductor

$$I = 170 \text{ A}$$

$|B| = ?$ in radial distance (a) $r = 0$



Fig 29-67

Ampere's law for B .
outside wire:

(b) $r = 1 \text{ cm}$

(c) 2 cm (wire surface)

(d) $4 \text{ cm} = r$

$$B = \frac{\mu_0 I}{2\pi r}$$

Inside wire: $B = \frac{\mu_0 I}{2\pi R^2} r$ $r = 0$

(a) $B = \left(\frac{\mu_0 I}{2\pi R^2} \right) r = 0$ Inside wire

(b) $B = \left(\frac{\mu_0 I}{2\pi R^2} \right) r = \frac{4\pi \times 10^{-7} \times 170 \times 1 \times 10^{-2}}{2\pi \times (2 \times 10^{-2})^2}$ Inside wire
 $B = 8.5 \times 10^{-4} \text{ T}$

(c) $r = a$; $B = \frac{\mu_0 I r}{2\pi r^2} = \frac{\mu_0 I}{2\pi r} = \frac{4\pi \times 10^{-7} \times 170}{2\pi \times (2 \times 10^{-2})}$

$$B = 1.7 \times 10^{-3} \text{ T}$$

Both formulas are correct to use in (c) part. That is, inside the wire or out the wire

(d) use formula outside the wire

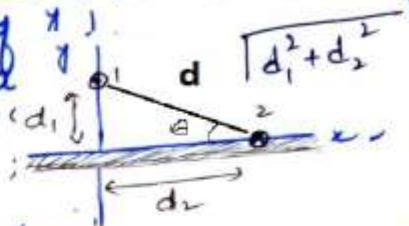
because $r > a$

outside wire: ~

$$B = \frac{\mu_0 I}{2\pi r}$$

Force b/w Two parallel Currents:

P-35: Fig 29-63 shows wire 1 in cross section; the wire is long and straight, carries a current of 4mA out of the page, and is at distance $d_1 = 2.4\text{cm}$ from a surface. wire 2, which is parallel to wire 1 and also long, is at horizontal distance $d_2 = 5\text{cm}$ from wire 1 and carries a current of 6.80mA into the page. What is the x-component of the magnetic force per unit length on wire 2 due to wire 1?



$i_1 = 4\text{mA}$ (current in long straight wire) out of the page

$d_1 = 2.4\text{cm}$ (wire 1)

$d_2 = 5\text{cm}$ (wire 2)

$i_2 = 6.8\text{mA}$ (into the page)

$F_x/L = ?$ (on the wire 2 due to wire 1)

Formula: $F = \frac{\mu_0 i_1 i_2 L}{2\pi d}$, $\sqrt{d_1^2 + d_2^2} = d$
 $\cos\theta = \frac{d_2}{\sqrt{d_1^2 + d_2^2}}$

So,

$$\frac{F_x}{L} = \frac{\mu_0 i_1 i_2 \cos\theta}{2\pi d}$$

$$= \frac{\mu_0 i_1 i_2}{2\pi d} \times \frac{d_2}{\sqrt{d_1^2 + d_2^2}}$$

$$\frac{F_x}{L} = \frac{\mu_0 i_1 i_2}{2\pi \sqrt{d_1^2 + d_2^2}} \times \frac{d_2}{\sqrt{d_1^2 + d_2^2}}$$

$$\frac{F_x}{L} = \frac{\mu_0 i_1 i_2 d_2}{2\pi (d_1^2 + d_2^2)} = \frac{4\pi \times 10^{-7} \times 4 \times 10^{-3} \times 6.8 \times 10^{-3} \times 5 \times 10^{-2}}{2\pi \times [(2.4 \times 10^{-2})^2 + (5 \times 10^{-2})^2]} = 8.84 \times 10^{-11} \text{ N/m}$$

P-1: A Surveyor is using a magnetic compass 6.1m below a power line in which there is a steady current of 100A. (a) what is the magnetic field at the site of the compass due to the power line? (b) will this field interfere seriously with the compass reading?

The horizontal component of Earth magnetic field at the site $20\mu T$?

$$i = 100A, r = 6.1m$$

(a) $\vec{B} = ?$ at the site of compass due to power line?

(b) Will this field interfere seriously with the compass Needle?

horizontal component of $B = 20\mu T$ at the site is

$$B = \frac{\mu_0 i}{2\pi r}$$

$$B = \frac{4\pi \times 10^{-7} \times 100}{2 \times 3.14 \times 6.1} = 3.3\mu T$$

Earth $B = 50\mu T$ but we'll use $B = 20\mu T$ (given)

✓ (b) $B = 3.3\mu T$ is $1/6^{th}$ of $B = 20\mu T$
Therefore it may affect the Compass Reading.

P.51: A 200 turn solenoid having length of 25cm and a diameter of 10cm carries a current of 0.29A. Calculate the Magnitude of the Magnetic field $\vec{B}$ inside the solenoid.

$$N = 200 \text{ (Solenoid)}$$

$$\text{Length} = 25\text{cm and diameter} = 10\text{cm}; i = 0.29\text{A}$$

$$B = ? \text{ (inside the Solenoid)}$$

$$B = \mu_0 i n = \mu_0 i \left(\frac{N}{l} \right) = 4\pi \times 10^{-7} \times 0.29 \times \left(\frac{200}{2.5 \times 10^{-2}} \right)$$

$$B = 3 \times 10^{-4} \text{ T}$$

P.52: A solenoid 1.30m long and 2.60cm in diameter carries a current of 18A. The Magnetic field inside the solenoid is 23.0 mT. Find the length of the wire forming the solenoid?

$$\text{Length of Solenoid} = 1.3\text{m}$$

$$\text{diameter} = 2.6\text{cm}; r = 1.3\text{cm}$$

$$i = 18\text{A}; B = 23\text{mT}$$

$$\text{Length of the wire forming the Solenoid} = ?$$

$$B = \mu_0 i n = \mu_0 i \frac{N}{l}; B = \mu_0 i \frac{N}{l}$$

$$\frac{B l}{\mu_0 i} = N \quad (1)$$

$$(2\pi r) N \text{ (loop} \times \text{No. of loops (turns))} = 2\pi r N$$

from eq (1)

$$= 2\pi r \frac{B l}{\mu_0 i}$$

$$= \frac{2\pi \times 1.3 \times 10^{-2} \times 23 \times 10^{-3} \times 1.3}{4\pi \times 10^{-7} \times 18} = \frac{3.88 \times 10^{-9}}{3.6 \times 10^{-6}}$$

$$\approx 108\text{m} \approx 107.7\text{m}$$

Fig 29-16a shows the cross-section of a long conducting cylinder with inner radius $a = 2\text{cm}$ and outer radius $b = 4\text{cm}$. The cylinder carries a current out of the page and magnitude of the current density in the cross section is given by $\vec{J} = cr^2$, with $c = 3 \times 10^6 \text{ A/m}^2$ and r in meters, what is the magnitude field $\vec{B}$ at the dot in Fig 29-16a, which is at radius $r = 3\text{cm}$ from the central axis of the cylinder?

We write the integral as

$$\begin{aligned} i_{\text{enc}} &= \int \vec{J} dA = \int_a^r cr^2 (2\pi r dr) \\ &= 2\pi c \int_a^r r^3 dr = 2\pi c \left[\frac{r^4}{4} \right]_a^r \\ i_{\text{enc}} &= \frac{\pi c (r^4 - a^4)}{2} \end{aligned}$$

We want the Magnetic field at the dot at radius r

$dA = 2\pi r dr$

Now we evaluate the left side of Ampere's law as we did in Fig 29-16 and we again obtain

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{\text{enc}}$$

$$B(2\pi r) = \frac{\mu_0 \pi c (r^4 - a^4)}{2}$$

Solving for B and substituting known data yield

$$\begin{aligned} B &= \frac{\mu_0 c}{4r} (r^4 - a^4) \\ &= \frac{(4\pi \times 10^{-7}) (3 \times 10^6)}{4(0.030)} \times ((0.030)^4 - (0.020)^4) \end{aligned}$$

Thus the mag. field $\vec{B}$ at a point 3cm from central axis has Magnitude $B = 2.0 \times 10^{-5} \text{ T}$